Unsupervised Machine Learning (lecture)

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February 3rd, 2026

Lecture 1:

Detection/Classification & Supervised Learning

*I have **so much data**… and just **manually annotated** some of it by category (species, call type, etc)!*

"Can I use my annotations to **detect and recognize** these categories in the rest of the data?"

"Can I use my annotations to **predict or estimate** additional information about these categories in my data?"

Lecture 2:

Unsupervised Learning

*I have **so much data**…how do I **learn more** about it?*

"Are there groups, or **clusters**, of similar-ish sounds in my data?"

"Are there unusual, or **anomalous**, sounds in my data?"

"How can I interpret the **overall structure** of my dataset?"

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What happens after detection?

What happens after detection?

Unsupervised Learning

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Key topics in unsupervised learning:

Dimensionality Reduction

Clustering

Anomaly Detection

What is "Dimensionality"?

The dimensionality of a dataset is the number of variables used to represent a data sample.

If we measure the number of cats each of you have:

0	2	1

1-D dataset

If we measure the number of cats and dogs each of you have:

2 0	0 15	1 1

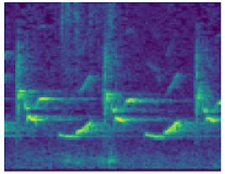
2-D dataset

If we open up a dataset that looks like this:

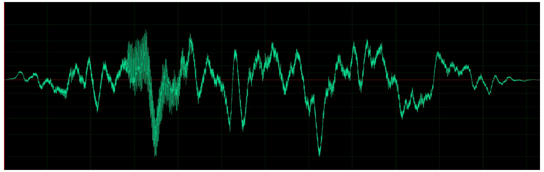
Subject	Period	Minf	Maxf	DF	CR	CD	CE	MT
1	before	456	2918	1850	12.67	0.128	1.88	-24
1	during	509	2838	1729	8.4	0.089	0.75	-24
1	after	452	3001	1773	13.76	0.131	1.81	-24

8-D dataset

What is the dimensionality of a spectrogram?



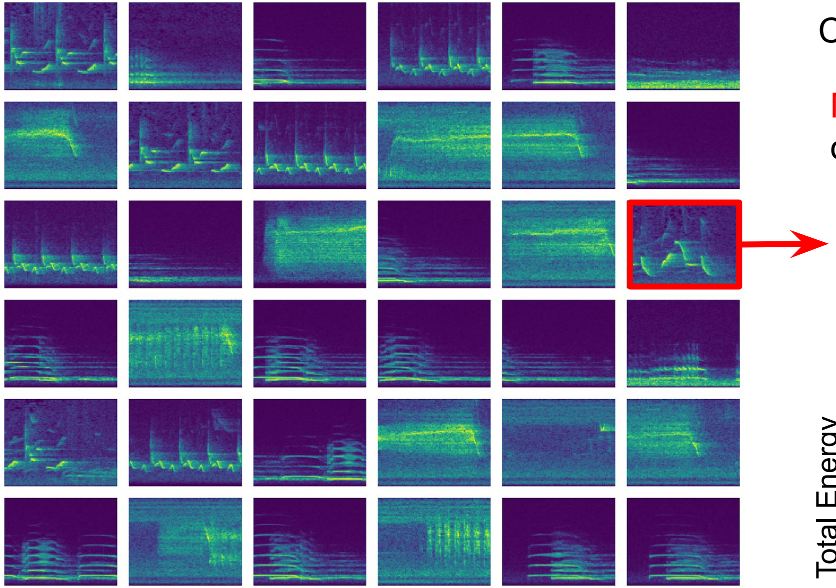
What is the dimensionality of a 10-second recording with a sampling rate of 48kHz?



High-dimensional data is difficult to work with, because we don't know which dimensions are most important and which are not. This brings us to **dimensionality reduction**!

Dimensionality Reduction

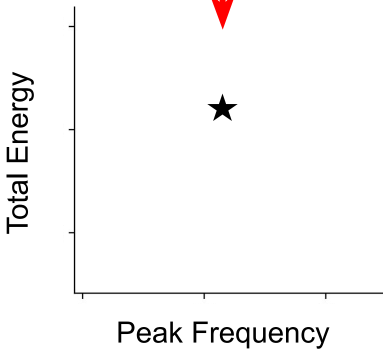
Dimensionality reduction is the problem of **transforming our data** from a **high-dimensional space** to a **low-dimensional space** while preserving the **meaningful characteristics** of the dataset.



Consider this dataset of 200 short spectrograms clips!

Manual feature extraction/selection can be thought of as one dimensionality reduction approach.

Features: duration, min/max/peak frequency, energy, etc.



Compare the "dimensions":

Spectrograms: *100 * 100 pixels = 10000*

Features: *2*

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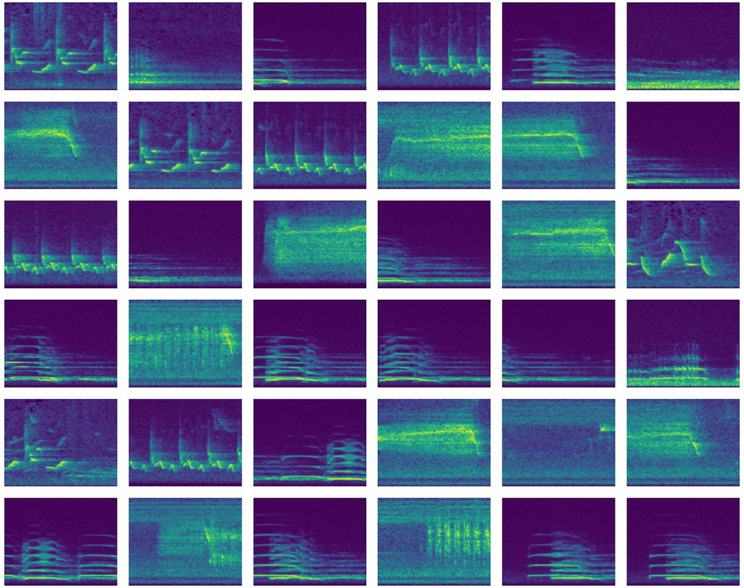
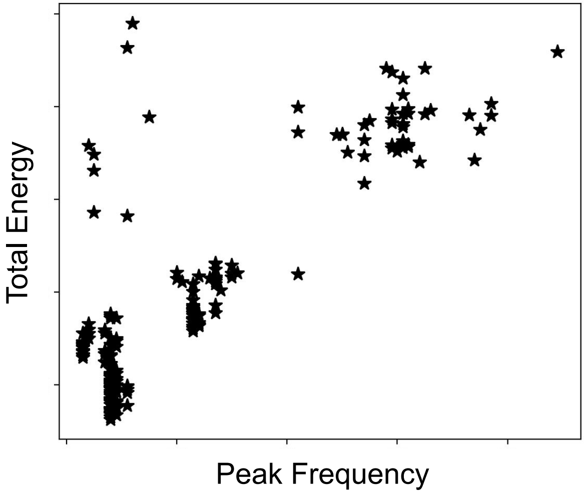


Let's calculate features for all points…



Instead of manually choosing features, can we use algorithms that **automatically extract** the **key characteristics** that vary in the dataset?

Dimensionality Reduction: Principal Component Analysis (PCA)

PCA is a foundational and incredibly useful method for dimensionality reduction.

input data = **principal components** **coefficients** **singular values**

(these define characteristics identified in the data)

(how much of each characteristic is in each data sample)

(how much each component contributes to variance in the dataset)

Each data sample will then be a linear combination of the “principal components”.

data sample = **component 1** × **coefficient 1** × **value 1** *(most significant characteristic)*

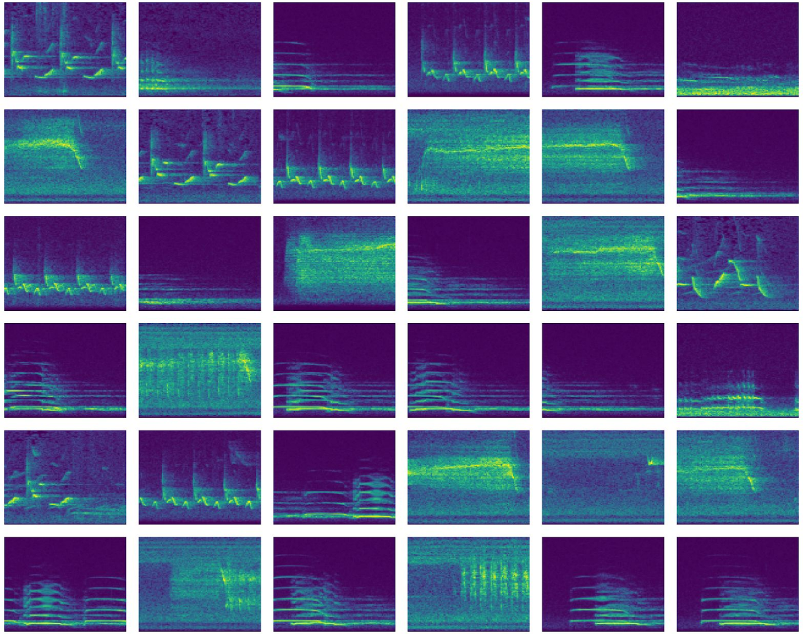
\+ **component 2** × **coefficient 2** × **value 2** *(second most significant characteristic)*

\+ **component 3** × **coefficient 3** × **value 3** + …

Dimensionality Reduction: Principal Component Analysis (PCA)

PCA can be applied to a feature vector to extract the most important features, or even directly to raw data!

input data



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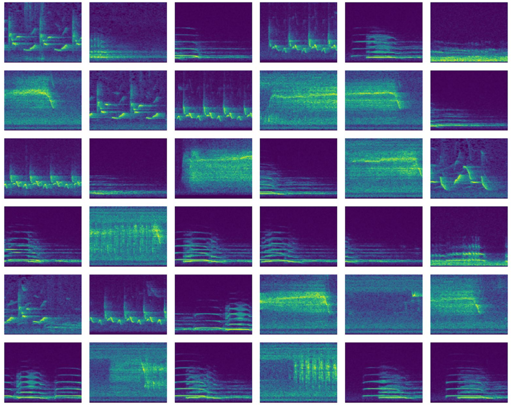
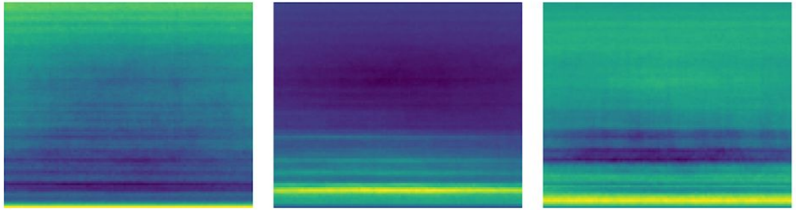
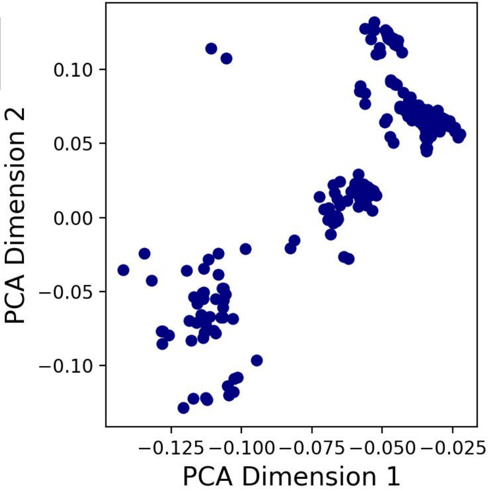


principal components



coefficients



singular values

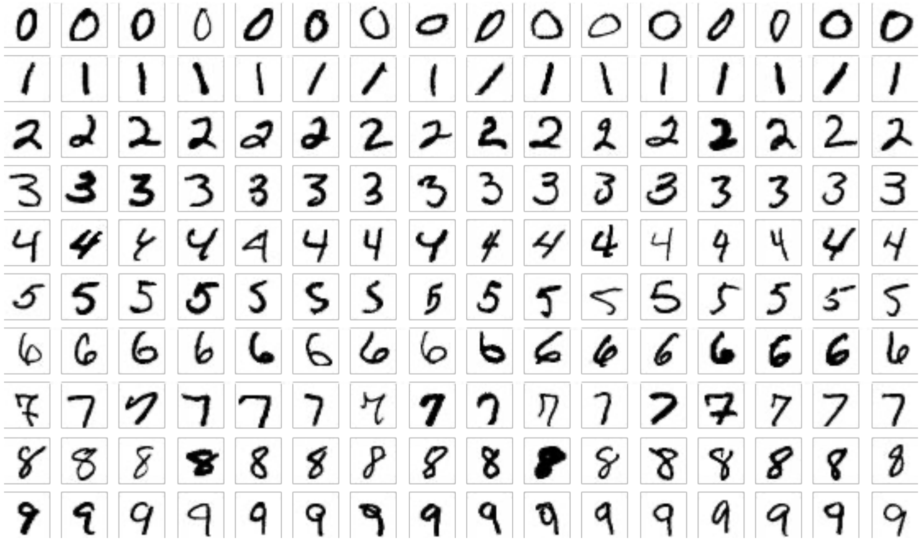
PC1: 30%
PC2: 10%
PC3: 5%

Dimensionality Reduction: Nonlinear Transformations

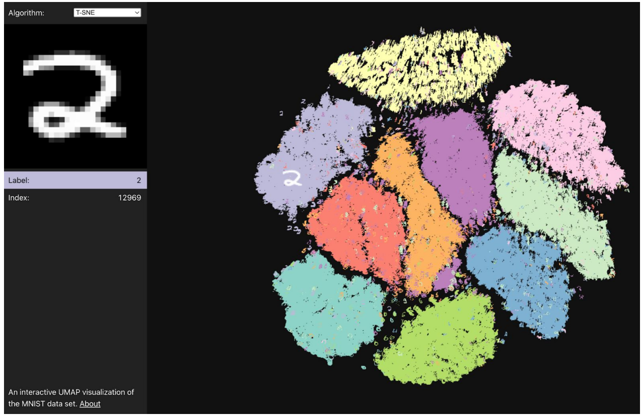
t-SNE (t-distributed Stochastic Neighbor Embedding): a **probabilistic** method for dimensionality reduction.

UMAP (Uniform Manifold Approximation and Projection): a **graph-based** method for dimensionality reduction.

Advantages (vs PCA): can represent nonlinear similarity in data.
Disadvantages (vs PCA): difficult to interpret the dimensions.

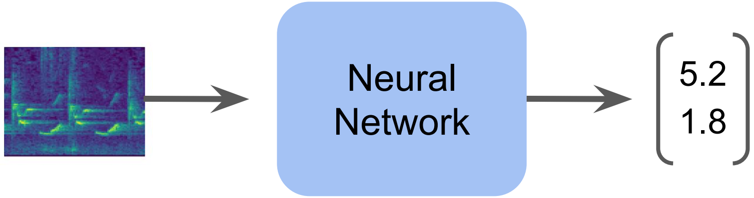


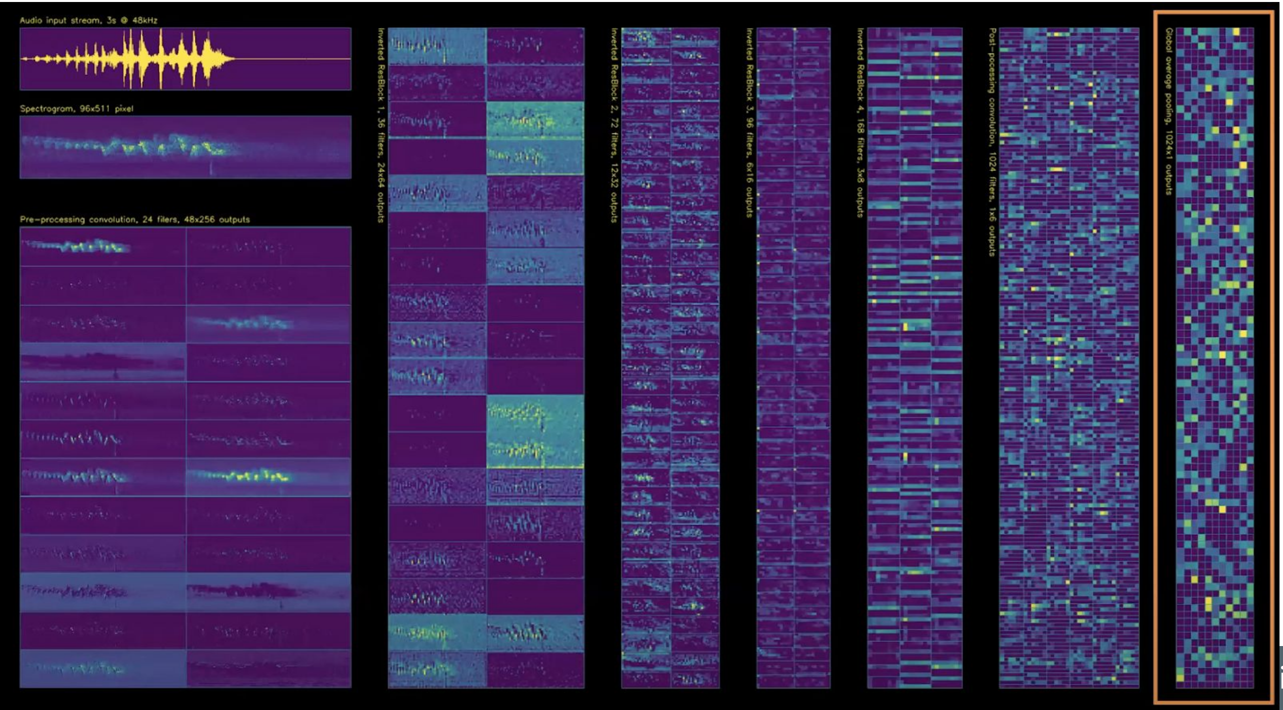
For visualization:
https://grantcuster.github.io/umap-explorer/



Dimensionality Reduction: Deep Learning and Embeddings

More recently, deep learning has also been used to perform dimensionality reduction. The lower-dimensional representation is usually called an **embedding**.

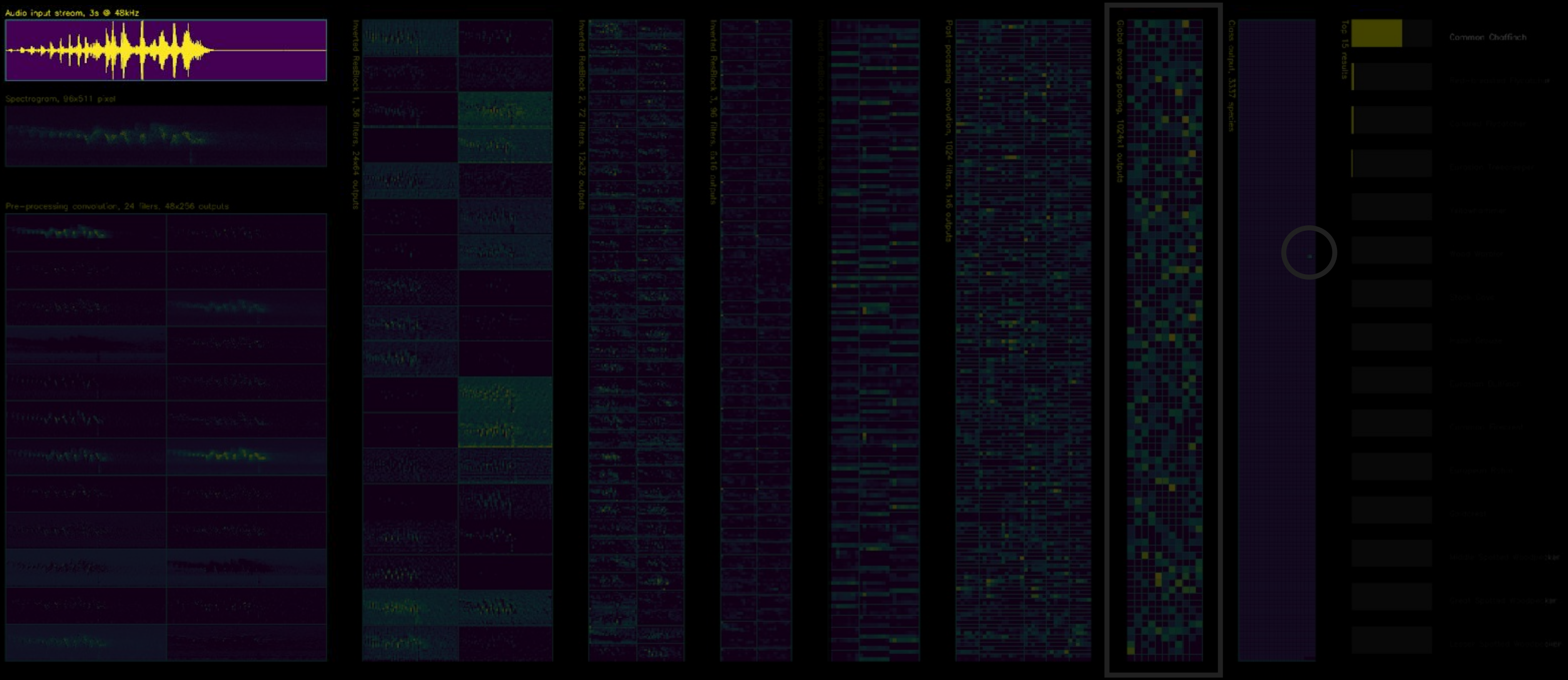




Recent work by the BirdNET Team has shown some really powerful uses of embeddings for bioacoustics!

Figure by Stefan Kahl.

BirdNET

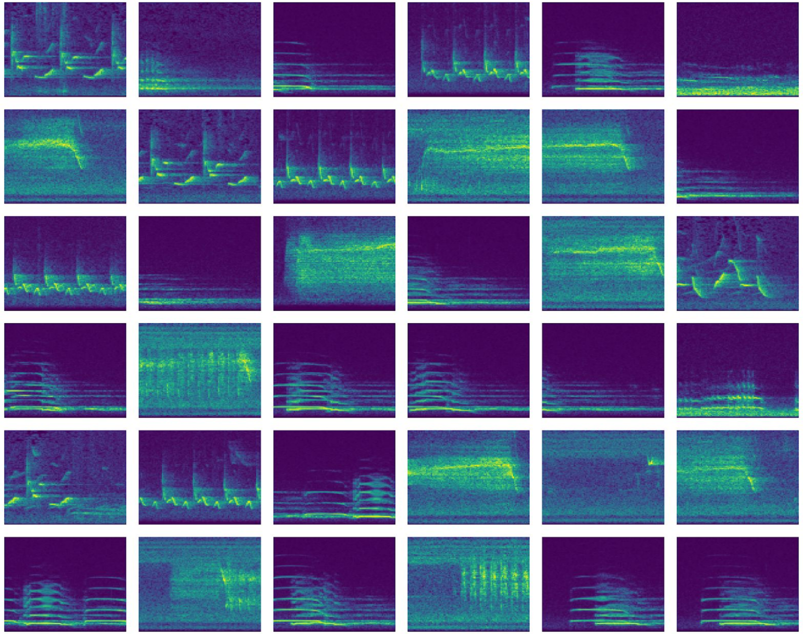


birdnet.cornell.edu

Dimensionality Reduction: t-SNE and UMAP

Let's apply t-SNE and UMAP to our spectrogram dataset (with an embedding dimension of 2):

input data



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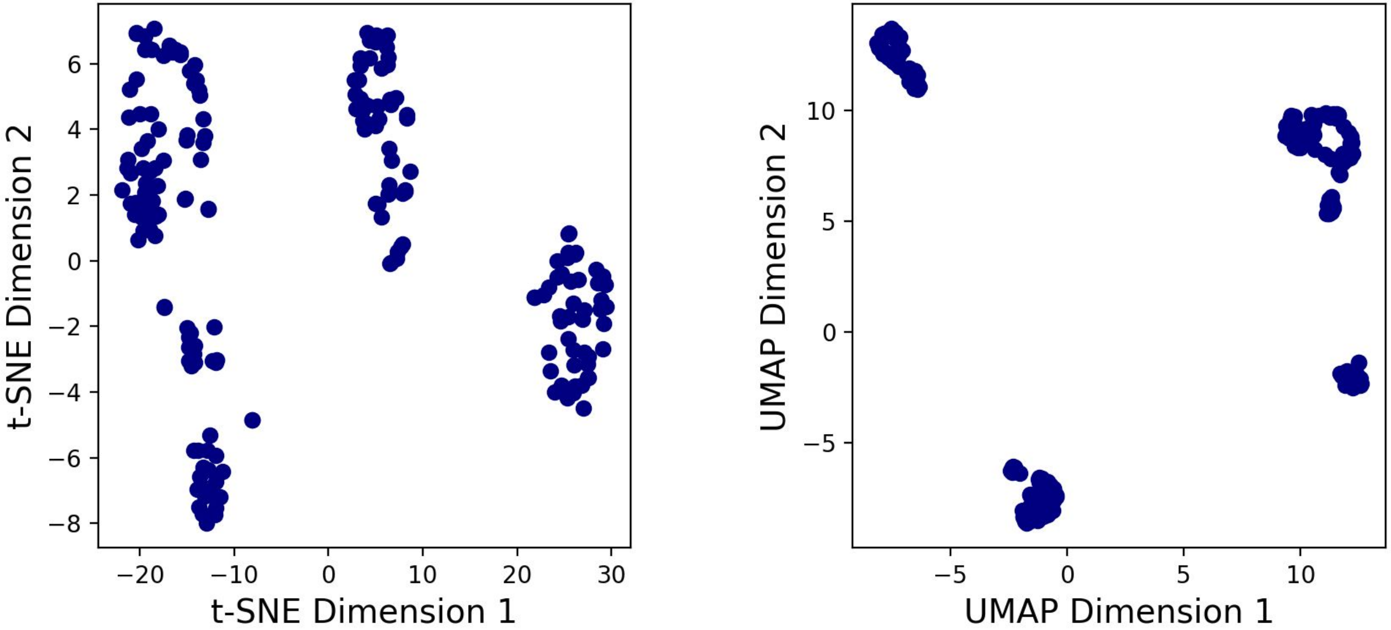

How might you interpret this output?

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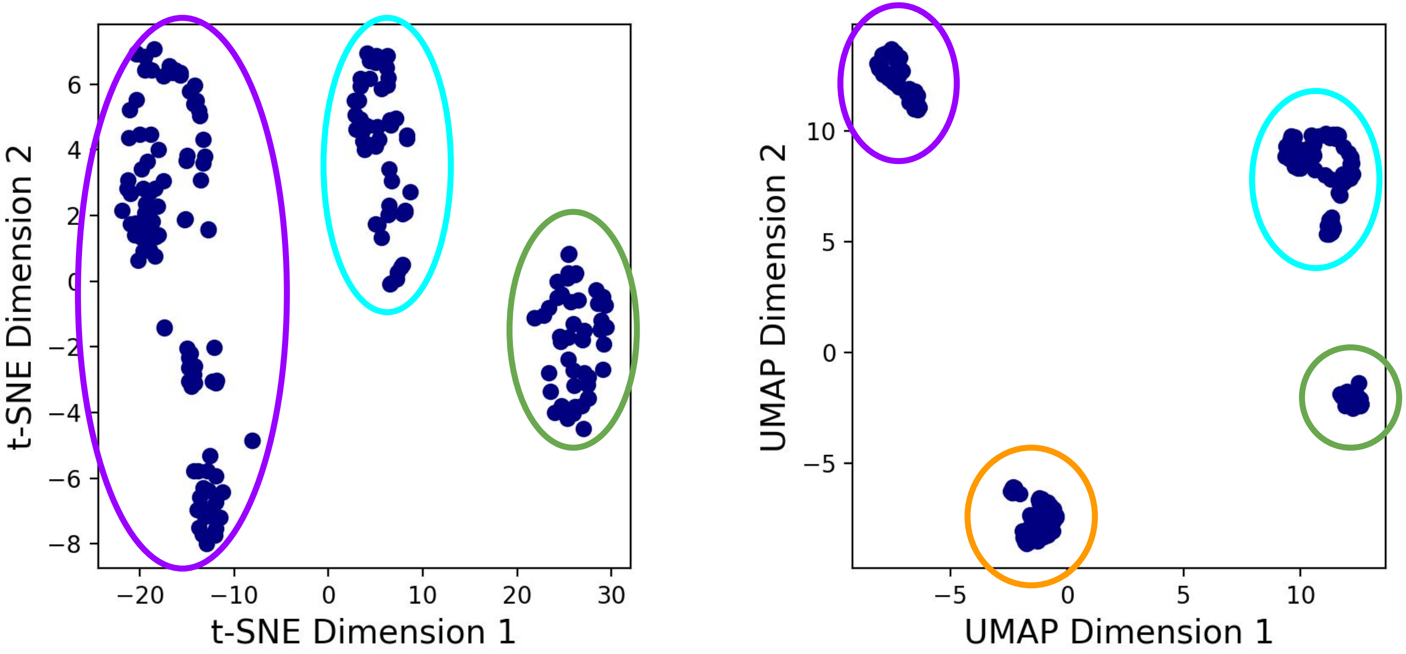


Is there **"structure"** in the dataset? For instance, are there **clusters**, and what do they represent?

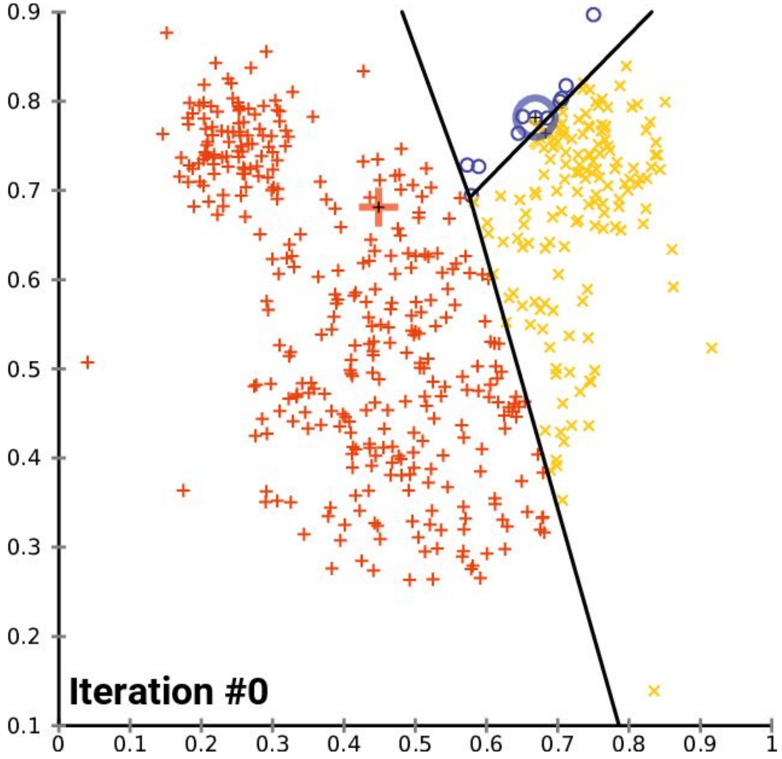
Clustering: K-Means

K-Means is a clustering algorithm that calculates **K "centroids"**. Each point is then associated with its nearest centroid.

On the right, K = 3 and the centroids are indicated by:

<u>Advantages:</u> simple and quick algorithm.

<u>Disadvantages:</u> assumes spherical clusters.



https://en.m.wikipedia.org/wiki/File:K-means_convergence.gif

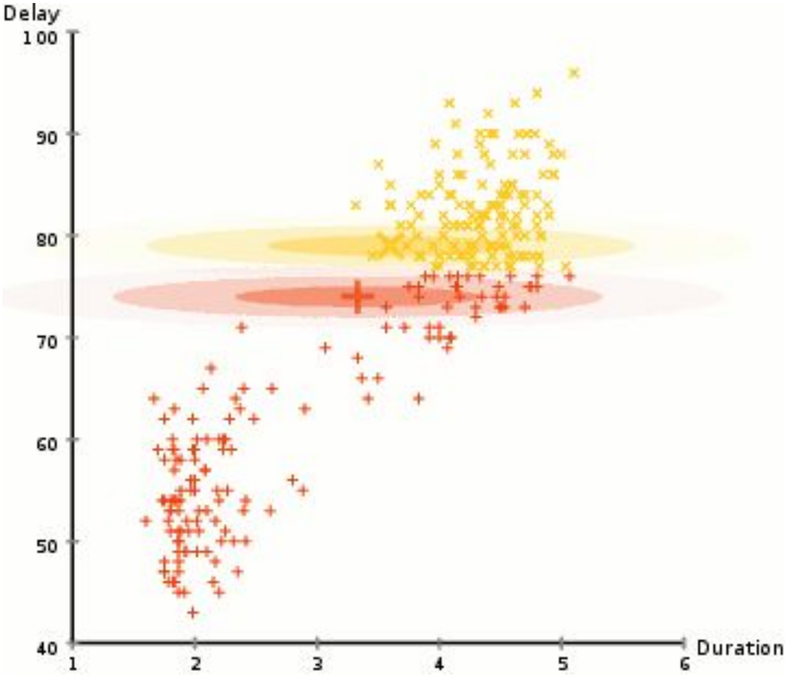
Clustering: Gaussian Mixture Model (GMM)

Gaussian mixture models the data as a **sum** of multiple **Gaussian** distributions.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

<u>Advantages:</u> points can be associated fractionally with multiple clusters.

<u>Disadvantages:</u> assumes Gaussian-distributed clusters.



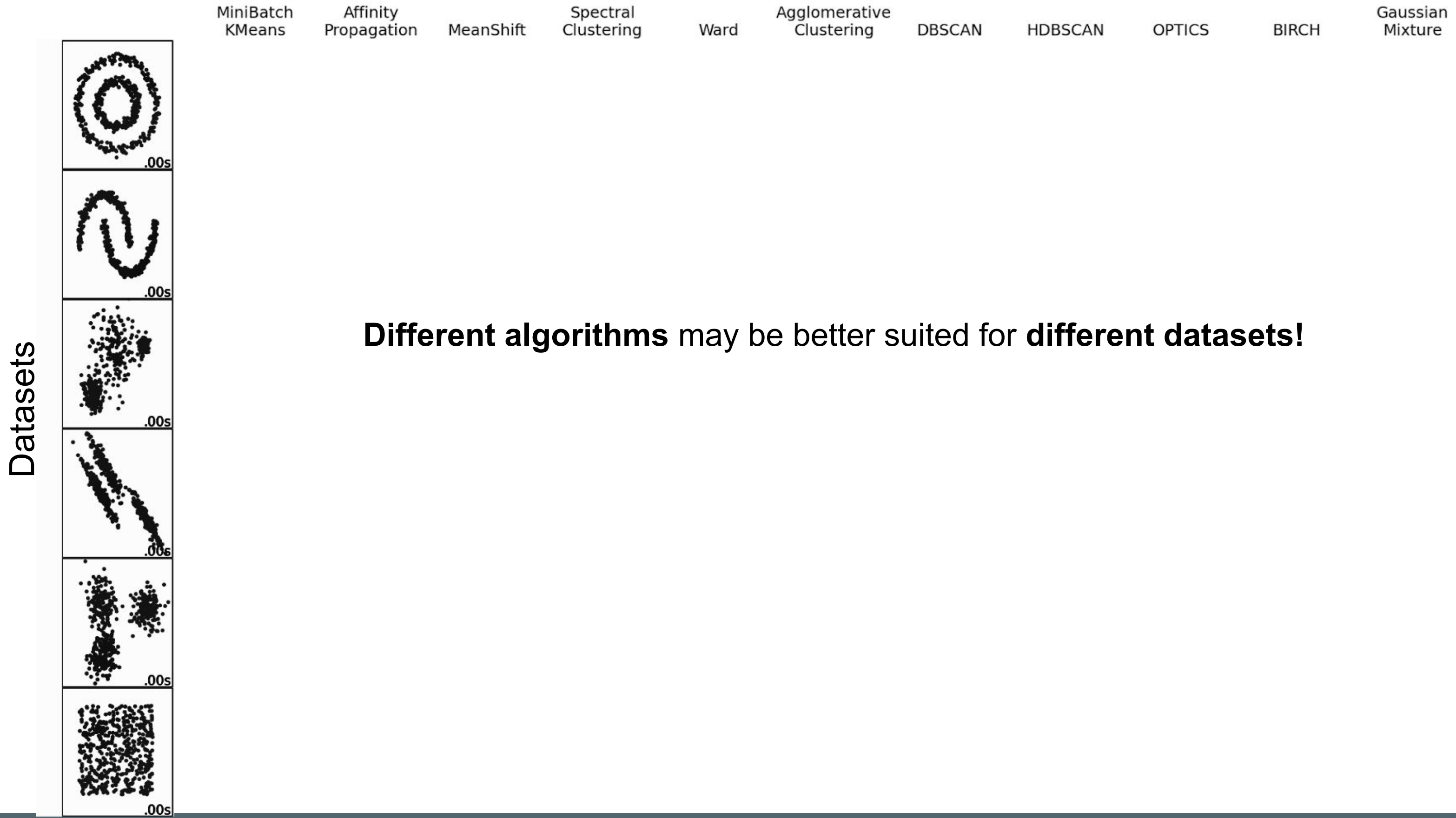
https://commons.wikimedia.org/wiki/FileEM_Clustering_of_Old_Faithful_data.gif

Other clustering methods…

MiniBatch KMeans	Affinity Propagation	MeanShift	Spectral Clustering	Ward	Agglomerative Clustering	DBSCAN	HDBSCAN	OPTICS	BIRCH	Gaussian Mixture

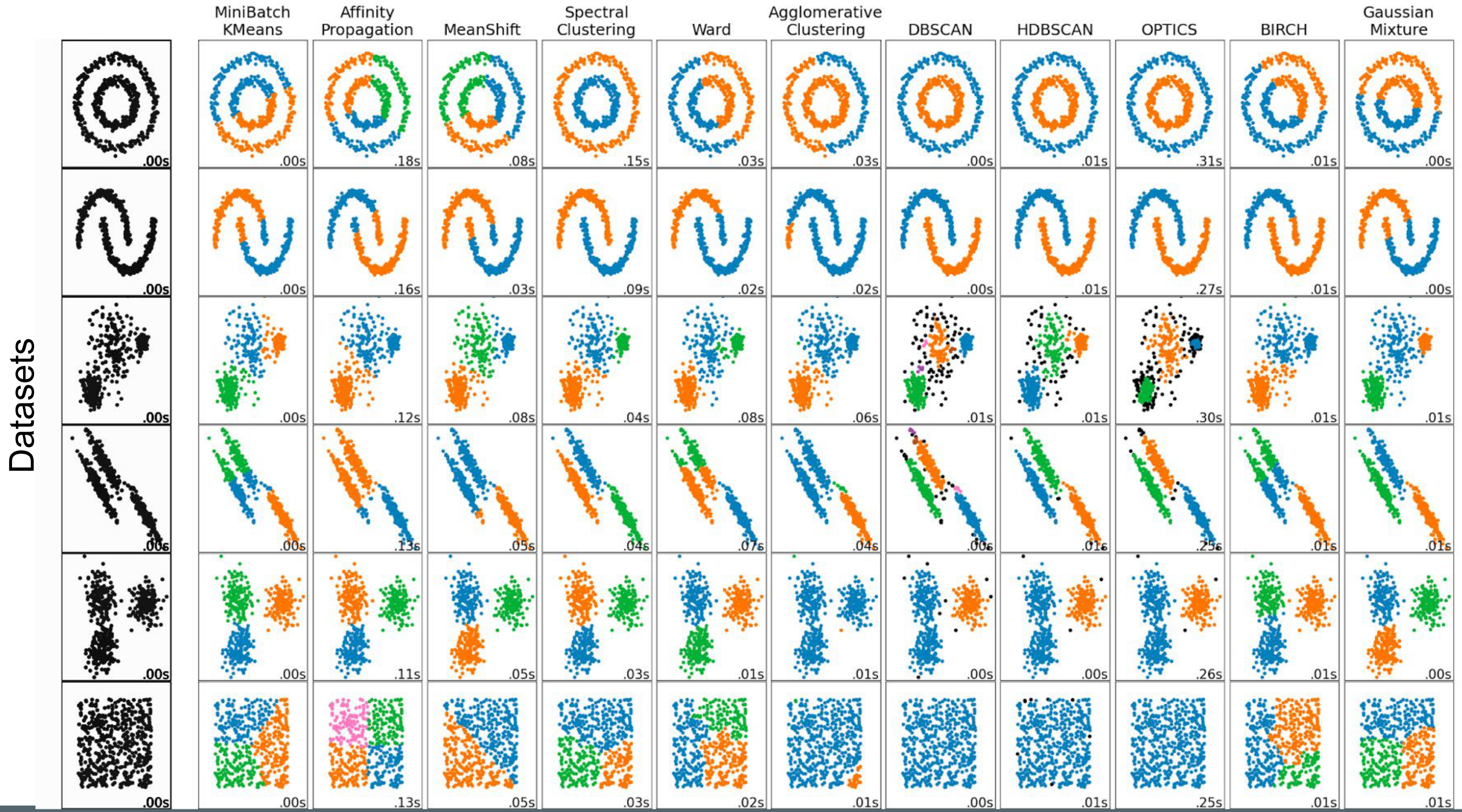
Why are there so many algorithms?

Other clustering methods…

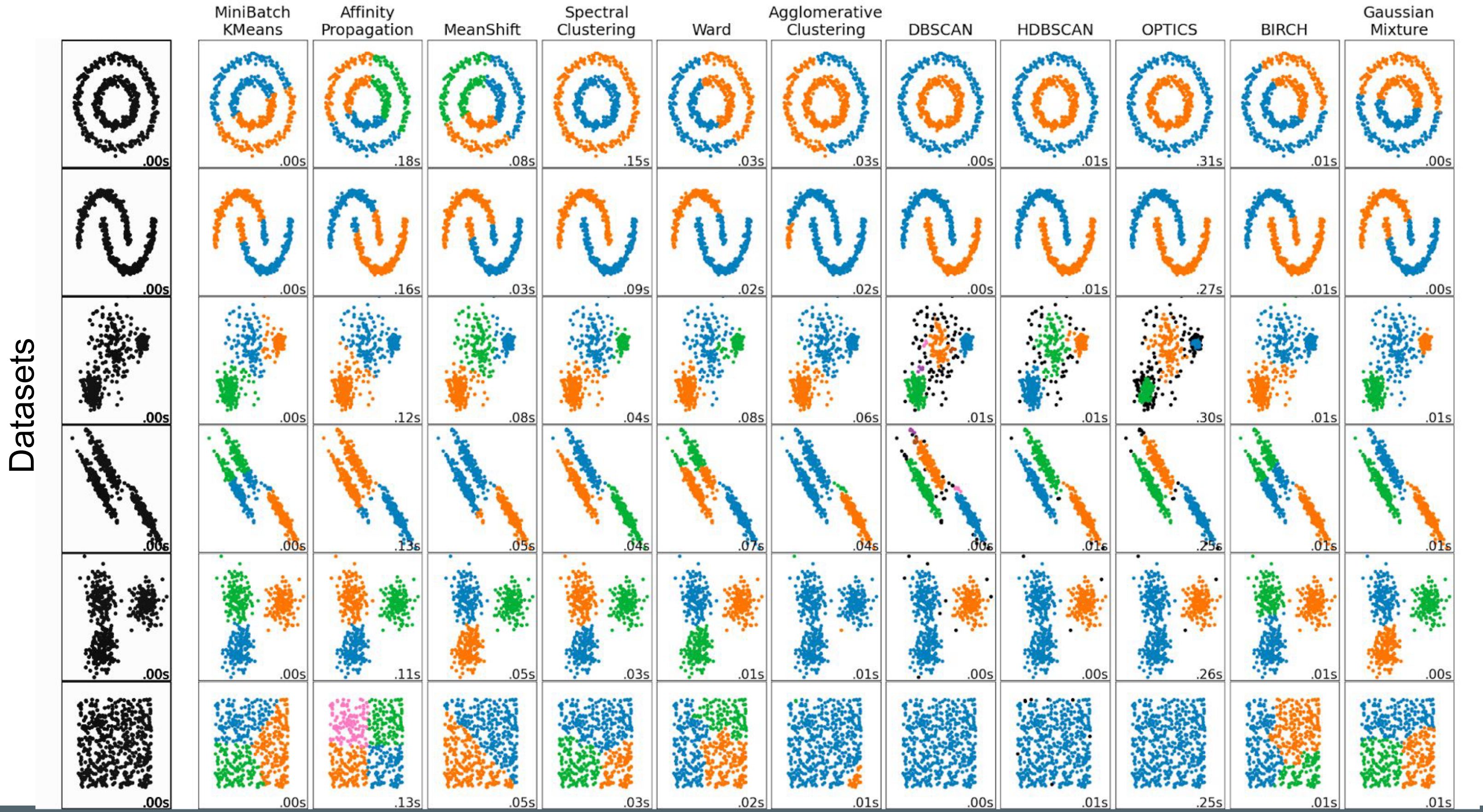


Different algorithms may be better suited for **different datasets!**

Other clustering methods…



For each dataset (row), what are the "best" clustering methods? Why?



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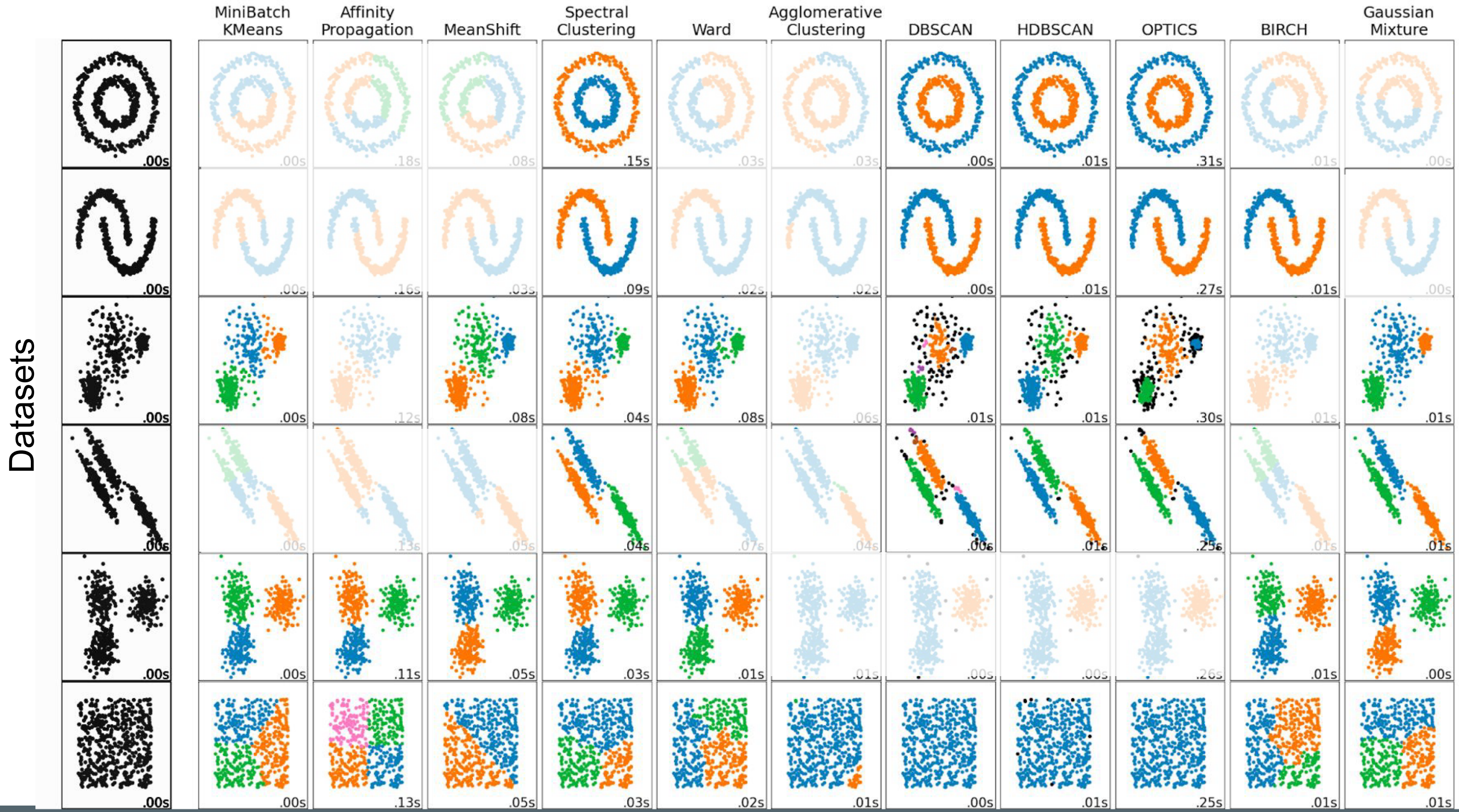
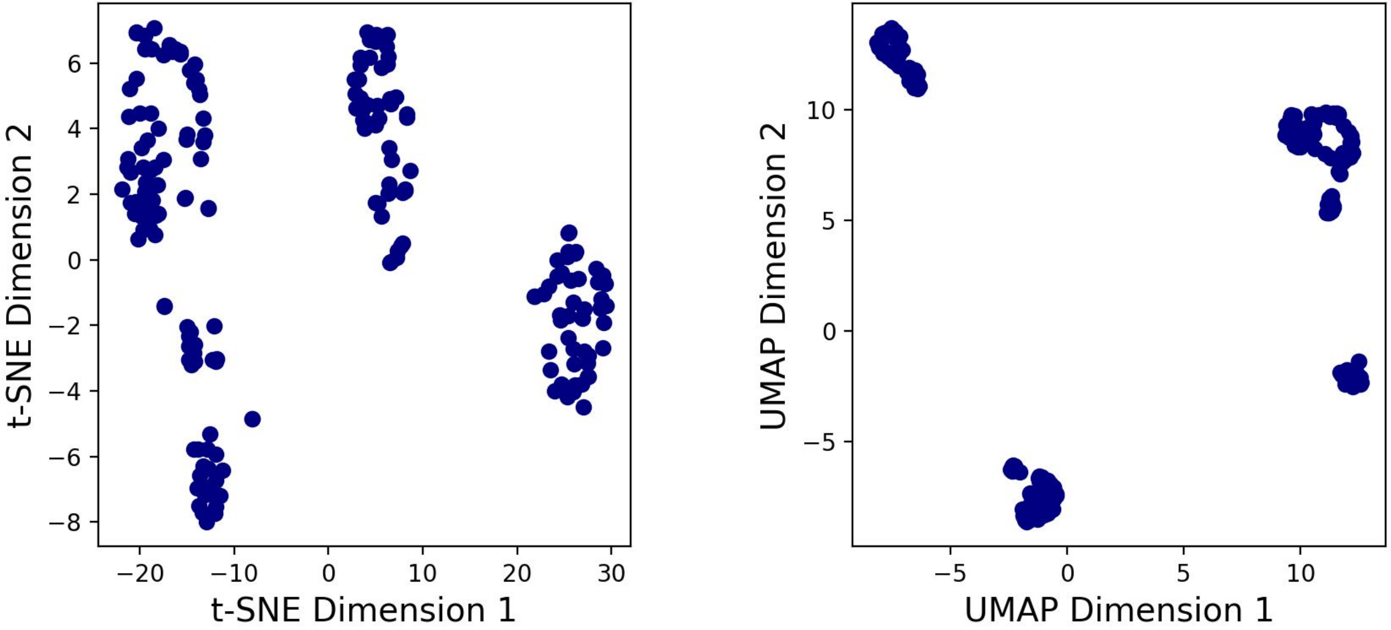


?

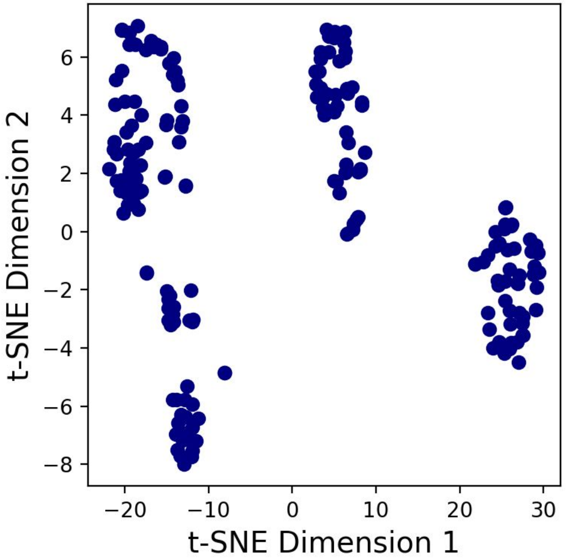
Back to our dataset…

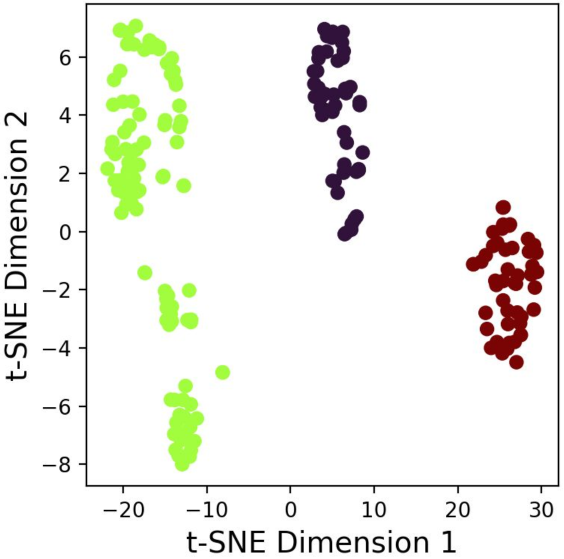
After applying t-SNE or UMAP to our datasets, let's cluster the points with k-means:

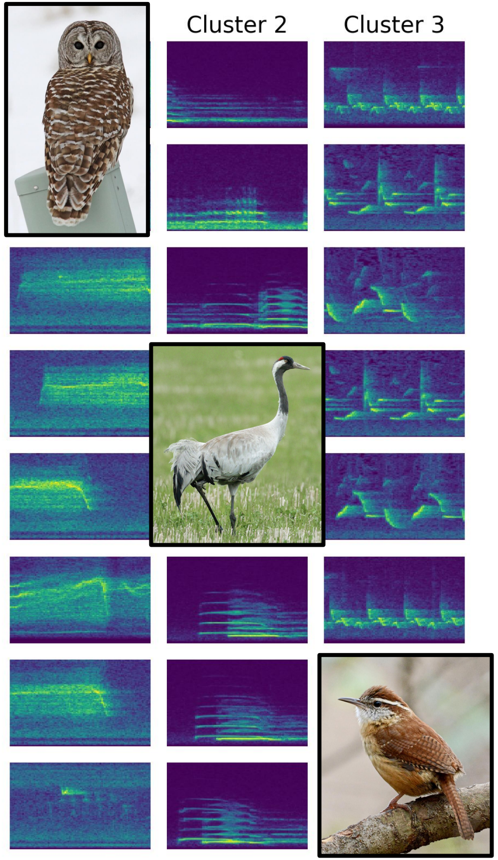


What should we choose for K?

Clustering: K-Means







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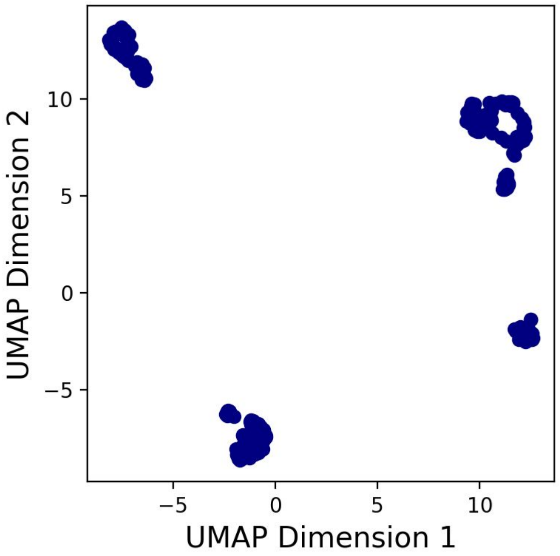
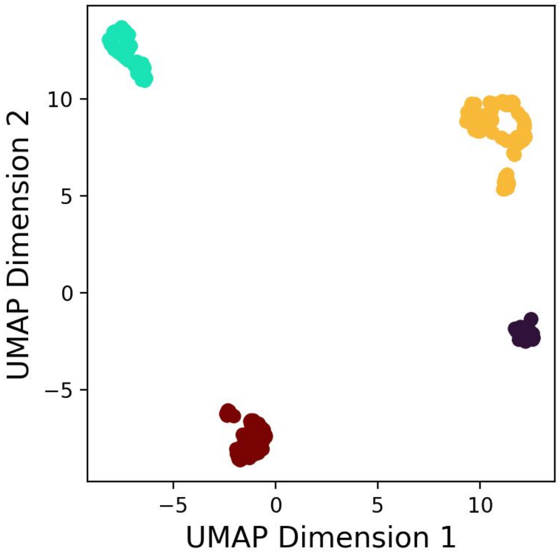
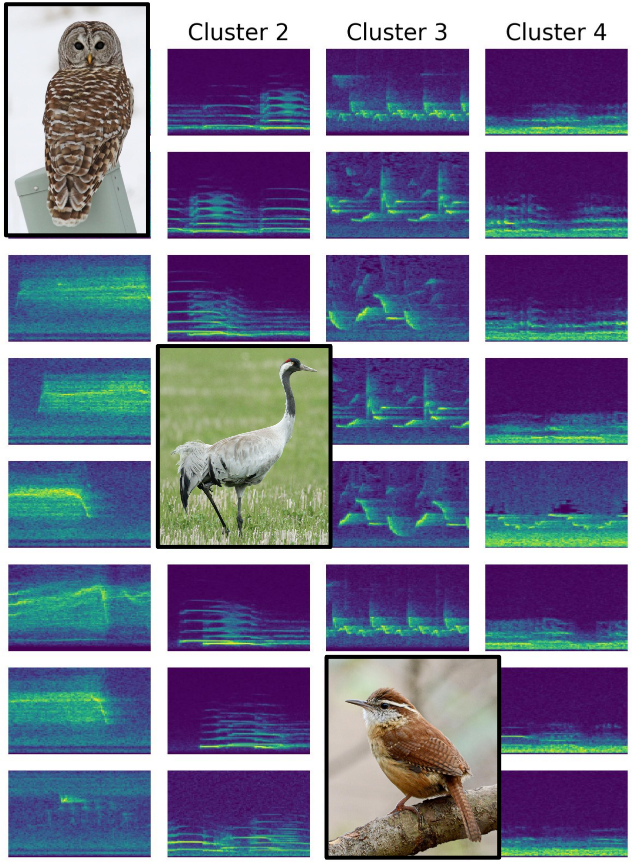


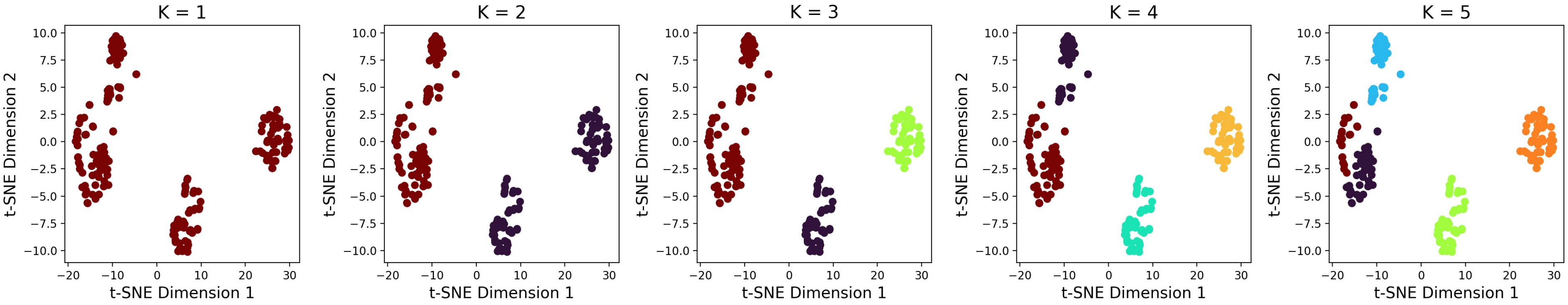


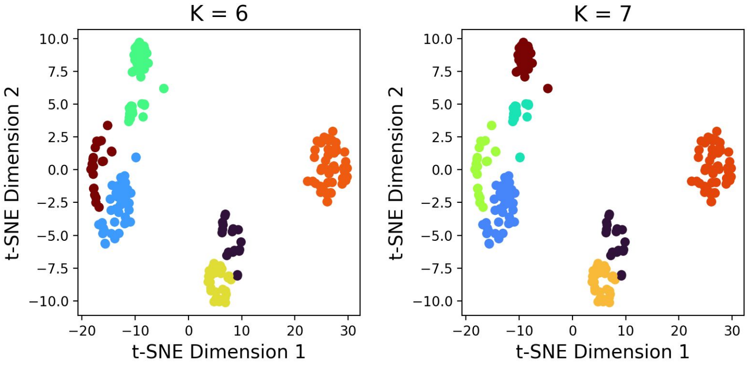


Clustering: Considerations

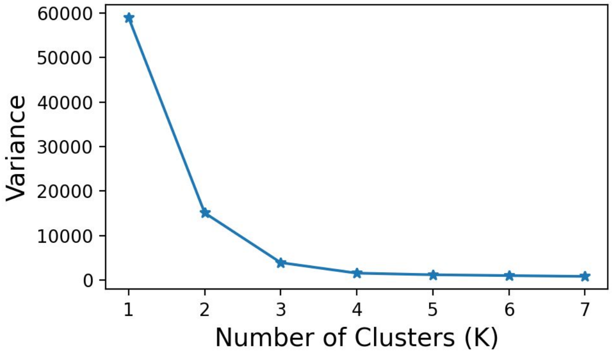
How to choose the **number of clusters**?

How to evaluate clustering **quality**?



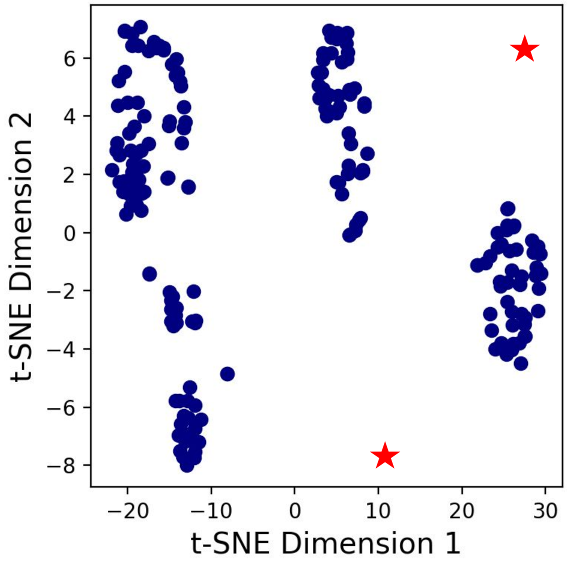


Elbow method: at which K does adding more clusters not really improve the "explained variation"?



Anomaly Detection

If we know what is **usual** or **typical** for our dataset, we can also find what is **unusual** or **atypical**.



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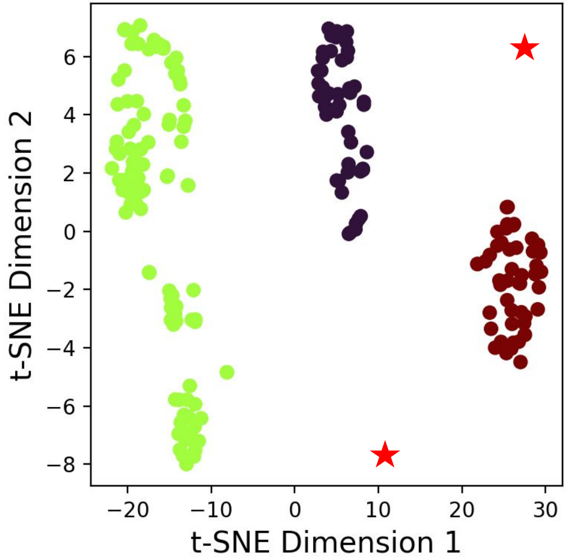


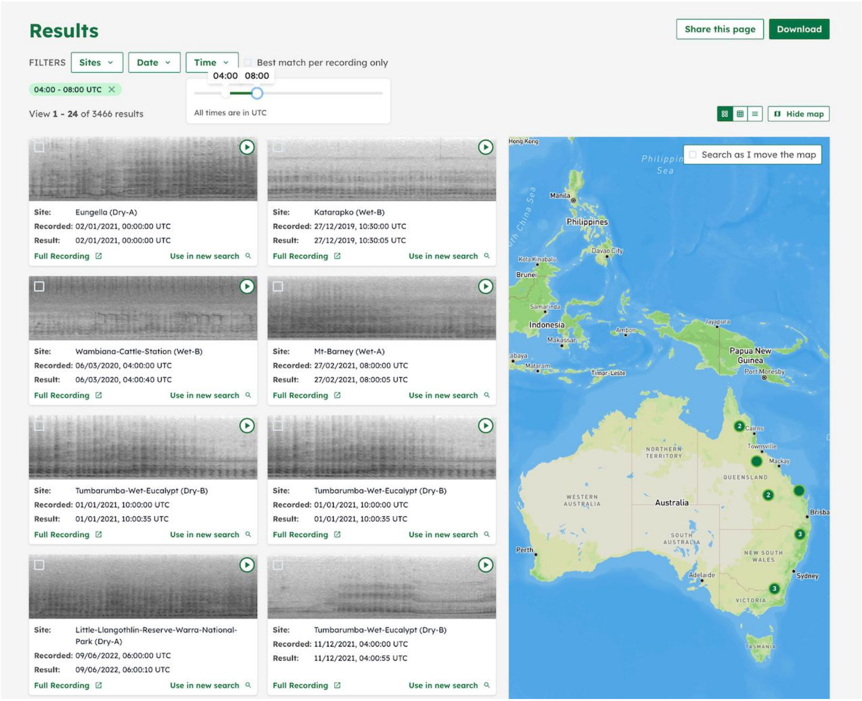
For example, points that are far from the clusters may be indicative of **anomalies** in the data.

Audio Search

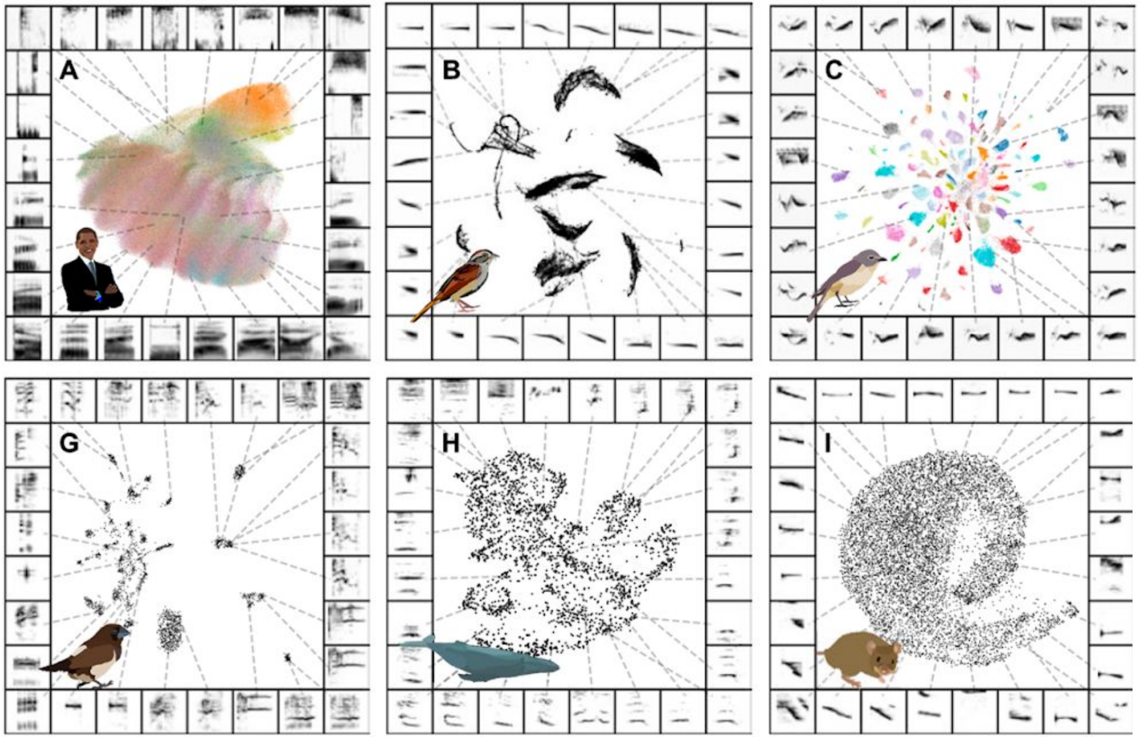
Imagine that you just have several, or even just one, example of a vocalization. How can you search an audio recording for similar sounds? This is termed *single-shot* or *few-shot learning.*

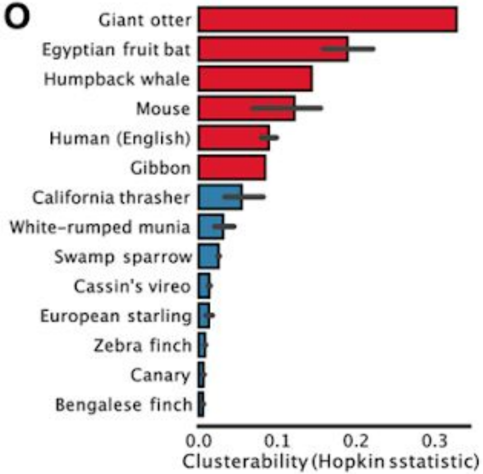
Australian Acoustic Observatory

https://search.acousticobservatory.org/



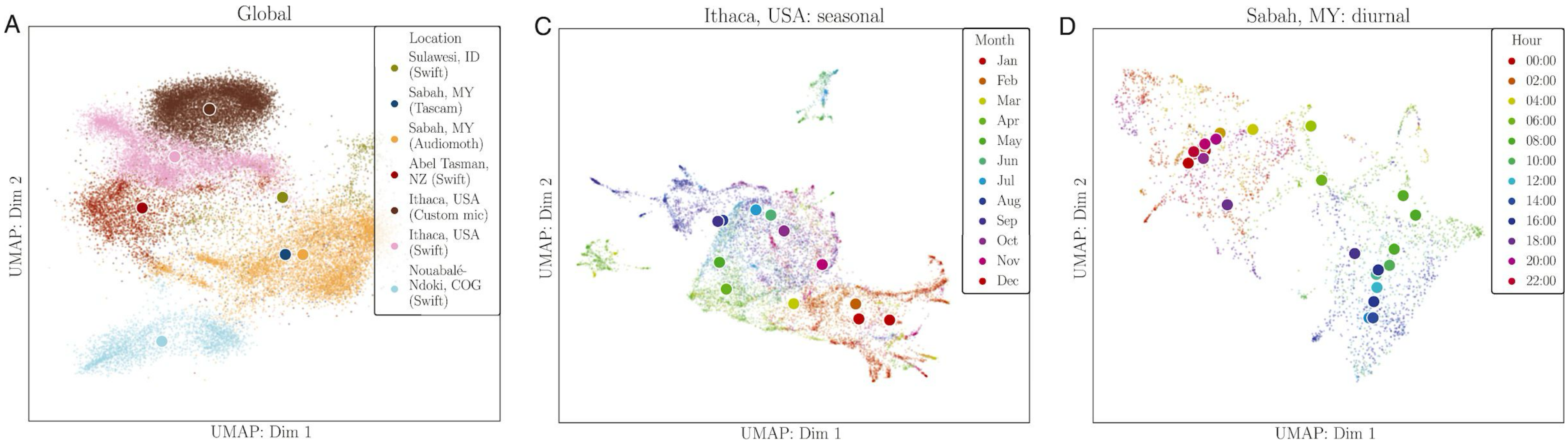
Suppose we are examining an audio dataset of **focal calls**. After performing dimensionality reduction, we see **clustering structure**. What might this clustering structure represent?





Sainburg, T., Thielk, M., & Gentner, T. Q. (2020). "Finding, visualizing, and quantifying latent structure across diverse animal vocal repertoires." *PLoS computational biology*, 16(10).

Suppose we are examining an audio dataset of **soundscapes.** After performing dimensionality reduction, we see **clustering structure**. What might this clustering structure represent?



Sethi, Sarab S., Nick S. Jones, Ben D. Fulcher, Lorenzo Picinali, Dena Jane Clink, Holger Klinck, C. David L. Orme, Peter H. Wrege, and Robert M. Ewers. "Characterizing soundscapes across diverse ecosystems using a universal acoustic feature set." Proceedings of the National Academy of Sciences 117, no. 29 (2020).

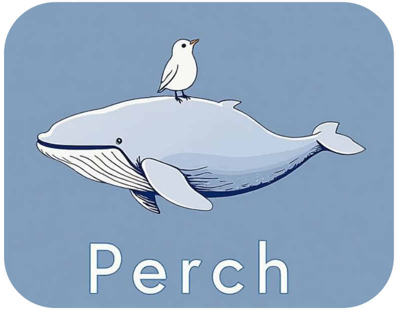
What does it mean to be a "good" embedding / feature vector?

Google DeepMind

How AI is helping advance the science of bioacoustics to save endangered species

Our new Perch model helps conservationists analyze audio faster to protect endangered species, from Hawaiian honeycreepers to coral reefs.

7 Aug 2025

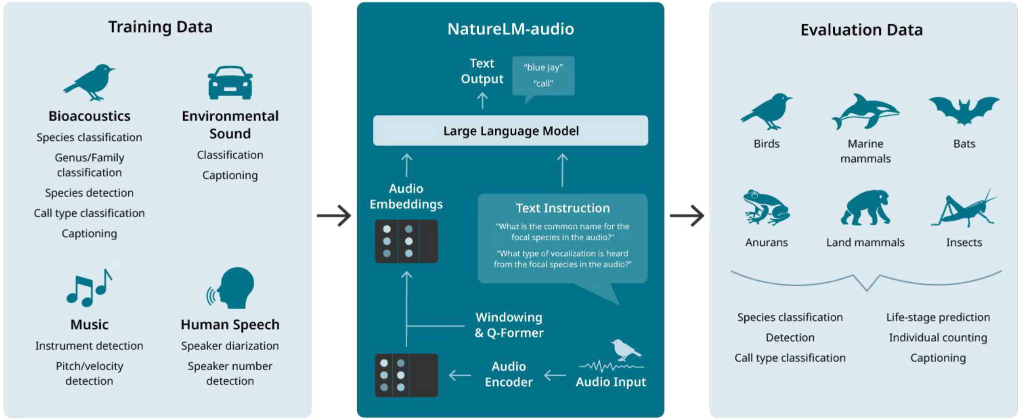


The Economic Times

What are animals saying? 3 AI tools that could soon tell us their thoughts

Scientists are using advanced AI tools to decode how animals communicate, with projects focused on dolphins, whales, and other species.

9 Jul 2025



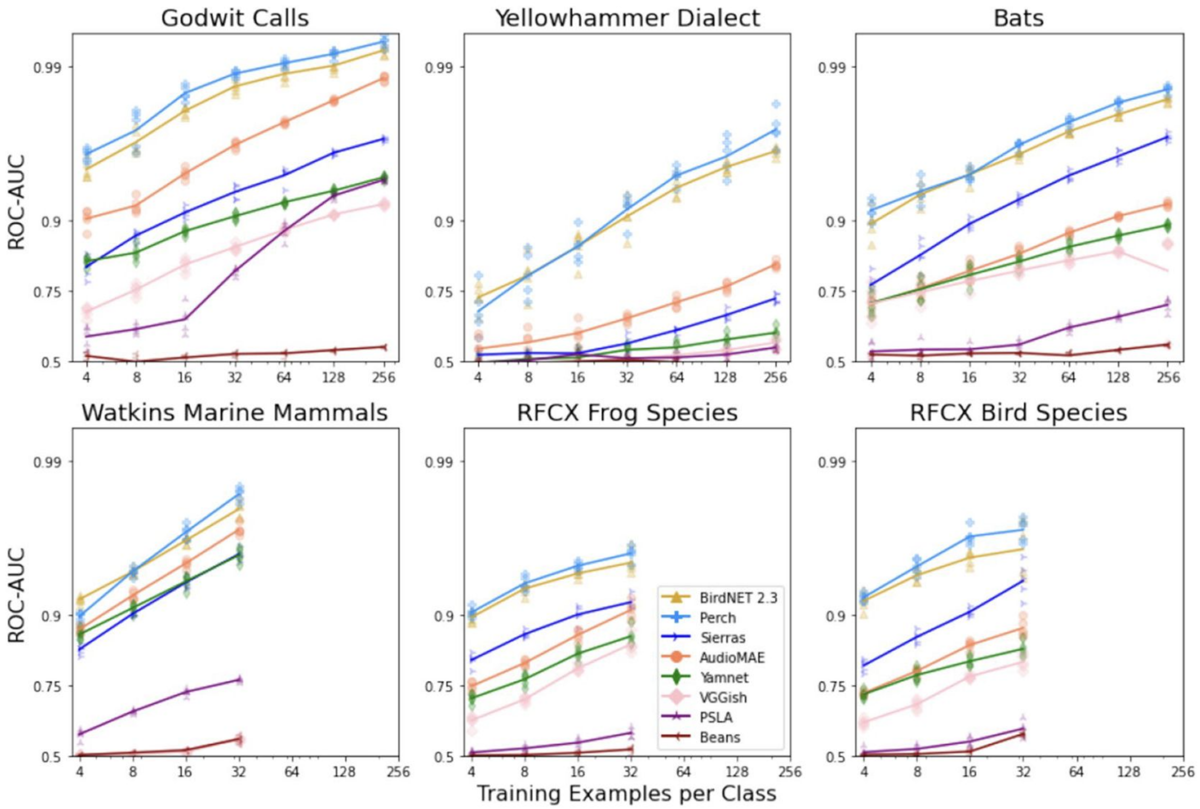
*"...foundation models: self-supervised models trained on **large amounts of unlabeled data** which can be **adapted to a variety of downstream tasks** with **minimal or no fine-tuning**…"*

Other Audio Embeddings

Model	Usage	General					Bioacoustic									
		AS-2M	AC	VGGS	FSD	ESC	MAC	XC	BS	INA	INS	MKT	ASA	WMM	RS	BEANS
Pure bioacoustic models																
Animal2Vec [76]	pretrain	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗
	eval	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓**0.91**	✗	✗	✗	✗
BirdMAE [85]	pretrain	✗	✗	✗	✗	✗	✗	✓	✓	✗	✗	✗	✗	✗	✗	✗
	eval	✗	✗	✗	✗	✓77.3	✗	✗	✓**44.0**	✗	✗	✗	✗	✗	✗	✗
BirdNET v2.4 [33]	pretrain	✗	✗	✗	✗	✗	✓	✓	✓[1]	✓	✗	✗	✗	✓	✗	✗
	eval	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
ConvNext$_{BS}$ [84]	pretrain	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗
	eval	✗	✗	✗	✗	✗	✗	✗	✓<u>36</u>	✗	✗	✗	✗	✗	✗	✗
Perch [60]	pretrain	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗
	eval	✗	✗	✗	✗	✗	✗	✗	✓<u>36</u>[2]	✗	✗	✗	✗	✗	✗	✗
ProtoCLR [72]	pretrain	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗
	eval	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
ViT$_{INS}$ [78]	pretrain	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓	✗	✗	✗	✗	✗
	eval	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓**60.3**	✗	✗	✗	✗	✗
Mixed-source models																
AVES [58]	pretrain	✓	✗	✓	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
	eval	✗	✗	✗	✗	✓77.3	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓52.8
BirdAVES [90]	pretrain	✓	✗	✓	✓	✗	✗	✓	✗	✓	✗	✗	✗	✗	✗	✗
	eval	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓55.1
BioLingual [75]	pretrain	✓	✓	✗	✗	✗	✗	✓	✗	✓	✗	✗	✓	✓	✗	✗
	eval	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓**83.8**
NatureLM-audio [86]	pretrain	✓	✓	✗	✗	✗	✗	✓	✓[1]	✓	✗	✓	✓	✓	✗	✗
	eval	✗	✗	✗	✗	✗	✗	✗	✓[3]	✗	✗	✗	✗	✗	✗	✓[3]
SurfPerch [77]	pretrain	✗	✗	✗	✓	✗	✗	✓	✗	✗	✗	✗	✗	✗	✓	✗
	eval	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗

Schwinger, Raphael, Paria Vali Zadeh, Lukas Rauch, Mats Kurz, Tom Hauschild, Sam Lapp, and Sven Tomforde. "Foundation Models for Bioacoustics – a Comparative Review." (2025)

Comparing Foundation Models

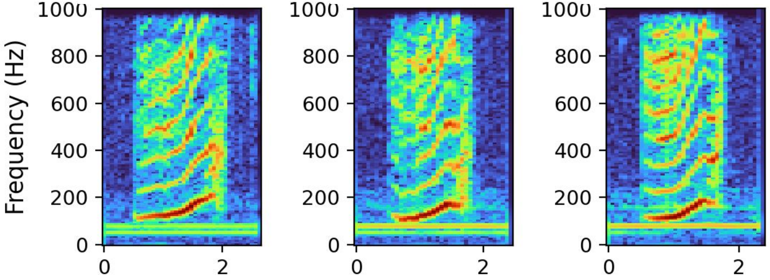


What performance do models achieve across a range of diverse “downstream” tasks?

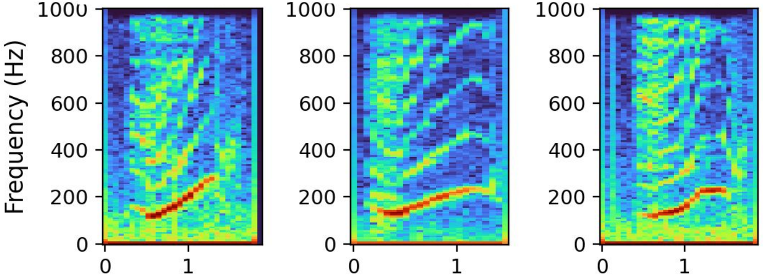
Ghani, Burooj, Tom Denton, Stefan Kahl, and Holger Klinck. "Global birdsong embeddings enable superior transfer learning for bioacoustic classification." *Scientific Reports* 13, no. 1 (2023):

Precautions with Audio Embeddings: Confounding Factors

Call Samples from Whale A



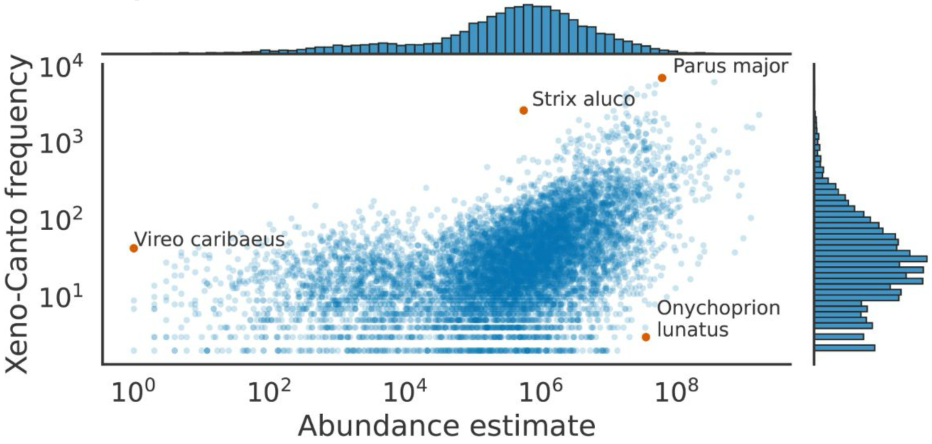
Call Samples from Whale B



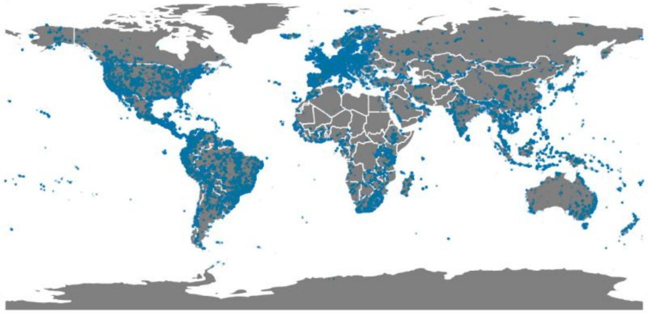
Precautions with Audio Embeddings: Bias

The **usefulness of features** learned by foundation models may **vary** with various factors, but their black-box nature may hide these effects.

Xeno-Canto data prevalence by species abundance:



Xeno-Canto data prevalence by geography:



Van Merriënboer, Bart, Jenny Hamer, Vincent Dumoulin, Eleni Triantafillou, and Tom Denton. "Birds, bats and beyond: Evaluating generalization in bioacoustics models." Frontiers in Bird Science 3 (2024)

To summarize…

Unsupervised Learning

*I have **so much data**…*

"Are there groups, or **clusters**, of similar-ish sounds in my data?"

"Are there unusual, or **anomalous**, sounds in my data?"

"How can I interpret the **overall structure** of my dataset?"

Key topics in unsupervised learning:

Dimensionality Reduction

Clustering

Anomaly Detection

Up next: unsupervised machine learning exercise!